

Data communications (Lab 12): Phase Modulation

Objective: To generate and analyze a Phase Modulated (PM) signal using MATLAB by varying the phase of a high-frequency carrier signal according to the amplitude of a low-frequency message signal.

Theory

In Phase Modulation (PM), the phase of the carrier signal changes according to the message signal. General equation: $s(t) = A_c \sin(2\pi f_c t + k_p m(t))$; where $m = A_m \sin(2\pi f_m t)$

where:

- $A_c = \text{carrier amplitude}$
- $f_c = \text{carrier frequency}$
- $m(t) = \text{message signal}$
- $k_p = \text{phase sensitivity}$
- $s(t) = \text{phase modulated signal}$

Main idea:

- Message signal positive $\rightarrow$ phase shifts forward
- Message signal negative $\rightarrow$ phase shift backward
- Amplitude remains constant

PM equation: $s(t) = A_c \sin(2\pi f_c t + k_p m(t))$;

Look at the angle inside sine: $2\pi f_c t + k_p m(t)$

- Normally carrier is: $A_c \sin 2\pi f_c t$
- Its phase is simply: $2\pi f_c t$
- In PM we add: $k_p m(t)$
- This extra term changes the angle and changing angle means changing **phase**.

How message changes Phase:

- Suppose: $m(t) = \sin(2\pi f_m t)$
- Message continuously varies: positive, zero, negative
- The value is multiplied by k_p and added to the carrier signal.

Case 1: Message positive

Example: $m(t) = +1$; then $x(t) = A_c \sin(2\pi f_c t + k_p)$

Added positive angle $\rightarrow$ wave shifts forward (left). This is called: Phase lead

Case 2: Message zero

Example: $m(t) = 0$; then $x(t) = A_c \sin(2\pi f_c t)$

No extra angle. No phase shifts. Wave is normal carrier.

Case 3: Message negative

Example: $m(t) = -1$; then $x(t) = A_c \sin(2\pi f_c t - k_p)$

Negative angle $\rightarrow$ wave shifts backward (right). This is called: Phase lag

Continuous change:

Message is sinusoidal: $m(t) = \sin(2\pi f_m t)$

So phase continuously changes:

positive $\rightarrow$ lead

zero $\rightarrow$ normal

negative $\rightarrow$ lag

zero $\rightarrow$ normal

positive $\rightarrow$ lead

again and again. That is why PM waveform looks compressed/expanded.

MATLAB Code

```
clear;
clc;
close all;

% Time vector
t = 0:0.0005:1;

% Message signal parameters
Am = 1;           % Message signal amplitude
fm = 5;           % Message signal frequency

% Generate message signal
m = Am*sin(2*pi*fm*t);

% Carrier signal parameters
Ac = 1;           % Carrier amplitude
fc = 50;          % Carrier frequency
kp = pi/2;        % Phase sensitivity

% Generate carrier signal
c = Ac*sin(2*pi*fc*t);

% Generate phase modulated signal
pm = Ac*sin(2*pi*fc*t + kp*m);

% Plot message signal
subplot(3,1,1)
plot(t,m)
grid on
title('Message Signal')
xlabel('Time (sec)')
ylabel('Amplitude')

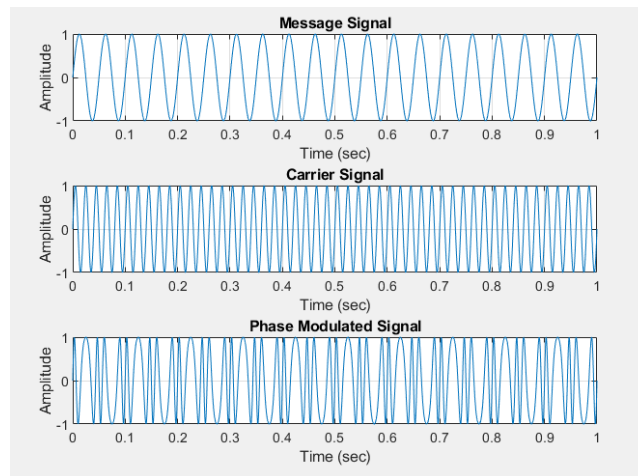
% Plot carrier signal
subplot(3,1,2)
plot(t,c)
grid on
```

```

title('Carrier Signal')
xlabel('Time (sec)')
ylabel('Amplitude')

% Plot phase modulated signal
subplot(3,1,3)
plot(t,pm)
grid on
title('Phase Modulated Signal')
xlabel('Time (sec)')
ylabel('Amplitude')

```



Code Logic

1. $m = A_m \sin(2\pi f_m t)$ → Generate the low-frequency message signal
2. $c = A_c \sin(2\pi f_c t)$ → Generate the high-frequency carrier signal
3. $p_m = A_c \sin(2\pi f_c t + k_p m(t))$ → This is the main PM equation; The message signal changes the phase of the carrier

Conclusion: The experiment successfully demonstrated Bipolar AMI Encoding using MATLAB. Binary 0 was represented by zero voltage, while binary 1 was represented by alternating positive and negative voltage levels. The output waveform verified the Alternate Mark Inversion principle.

Assignment: Write a MATLAB program to generate a Frequency Modulated (FM) signal using a message signal given as input by the user. The program should plot the message signal, carrier signal, and frequency modulated signal.